Supplementary Material for Real Analysis (2026)

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The material in this document will supplement the lectures by the instructor. Although this document will mostly be self-contained, from time to time the knowledge of some basic definitions and concepts will be assumed.

1 Defining the real numbers 1

1.1 Ordered fields 2
1.2 Least upper bounds 4
1.3 Exercises 6

2 Sequences 7

2.1 Convergent and Cauchy sequences 7
2.2 Limit superior and limit inferior 11
2.3 Exercises 13

3 Series 14

3.1 Convergence, absolute and conditional convergence 14
3.1.1 Exercises 17
3.2 Convergence of the series $\sum a_n b_n$ 18
3.3 Cauchy products of sequences and series 20
3.3.1 Exercises 22

4 Metric spaces and continuity 23

4.1 Metric Spaces 23
4.1.1 Exercises 23
4.2 Continuity 24
4.3 Uniform continuity 27
4.3.1 Exercises 29

1 Defining the real numbers

Question 1.1. What is the difference between an *operation* and a *relation*? Think of examples.

A *group* G is a set equipped with the binary operation $+$ (addition) such that

1. the operation $+$ is associative, that is, for all $a, b, c \in G$, we have $a + (b + c) = (a + b) + c$,
2. there is an (identity) element $0 \in G$ such that $a + 0 = a$ for all $a \in G$,
3. for every $a \in G$, there is an element $-a \in G$ such that $a + (-a) = 0$.

By definition, if G is a group then $|G| \geq 1$. There is no empty group.

A *field* F is a set equipped with binary operations $+$ and $\cdot$ such that

1. $(F, +)$ is an Abelian group (the identity is 0)
2. $(F - \{0\}, \cdot)$ is also an Abelian group (the identity is 1)
3. $\cdot$ distributes over $+$, that is, $a\,(b + c) = ab + ac$ for all $a, b, c \in F$.

Question 1.2. Why is there no field of cardinality 1?

Again, by definition it is clear that if F is a field then $|F| \geq 2$. A field has at least an additive and a multiplicative identity which are distinct. However, in many other equivalent formulations of the field definition the condition $|F| \geq 2$ is explicitly incorporated. In such formulations, the distinctness of the additive and multiplicative identity is inferred. In fact, if $0 = 1$, then the field necessarily collapses to the singleton $\{0\}$. To see this, note that in F

$$0a = 0 \quad \forall a. \qquad [\textit{Proof. } 0a = (0 + 0)\,a = 0a + 0a \Longrightarrow 0a = 0]$$

As a corollary, we get that if $0 = 1$, then $F = \{0\}$. In F we also have

$$(-1)\,a = -a \quad \forall a \quad [\textit{Proof. } (-1 + 1)\,a = 0 \Longrightarrow (-1)\,a = -a],$$

a small result which we will use in the next subsection.

1.1 Ordered fields

An *ordered field* is a field F equipped with a **total** order such that the field operations are compatible with the order in the following manner:

1. if $a \leq b$ then $a + c \leq b + c$ for all c,
2. if $a \leq b$ and $c \geq 0$ then $ac \leq bc$.

So, we have seen that given a set, one can give structure to it in multiple ways. One way is to assign operations on the set. In this direction we get groups, fields etc. Another is to give it an order, which gives rise to ordered sets. An ordered field combines both the structures. The following diagram displays this as a hierarchy diagram.

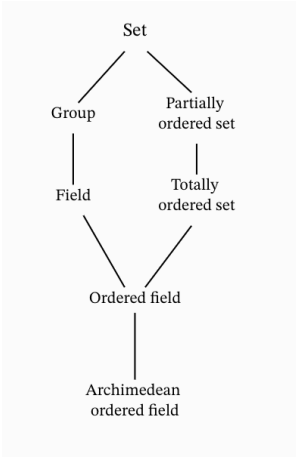


Figure 1.1. Hierarchy diagram of structured sets

We say an ordered field F is *Archimedean* if whenever $x, y \in F$ such that $0 < x < y$, then there exists $n \in \mathbb{N}$ such that $y < nx$. Here, in a general field by nx we mean x added to itself n times.

Informally, the Archimedean property says that no element of the field is either too large or too small. Some properties of ordered fields are listed below with their proofs.

$a \leq b \Longrightarrow -b \leq -a$	$a + (-b - a) \leq b + (-b - a) \Longrightarrow -b \leq -a.$
$a \leq b \Longrightarrow bc \leq ac \quad \forall c \leq 0$	$a \leq b \Longrightarrow a(-c) \leq b(-c) \Longrightarrow -ac \leq -bc \Longrightarrow bc \leq ac.$
$a \geq 0,\ b \geq 0 \Longrightarrow ab \geq 0$	$a \geq 0 \Longrightarrow ab \geq 0b \Longrightarrow ab \geq 0.$
$a^2 \geq 0 \quad \forall a$	Only need to show for when $a \leq 0$. In that case $-a \geq 0$. Then $a^2 = (-1)(-1)\,a^2 = (-a)(-a) \geq 0.$
$0 \leq 1$	$(-1)^2 = 1$
$a > 0 \Longrightarrow a^{-1} > 0$	If $a^{-1} < 0$ then $a^{-1}a < 0a \Longrightarrow 1 < 0$. Contradiction.
$0 \leq a \leq b \Longrightarrow 0 \leq b^{-1} \leq a^{-1}$	$a, b \geq 0$ then $a^{-1}b^{-1} \geq 0$. Multiply $a \leq b$ with $a^{-1}b^{-1}$.

Note that F being totally ordered is important in the proof of the statement $a^2 \geq 0$, and (therefore) for all the statements that follow from it.

1.2 Least upper bounds

An element $a \in W$ is said to be an *upper bound* for $S \subseteq W$ if $a \geq s$ for all $s \in S$. Now, a is called a *least upper bound* or an *supremum* for S if whenever b is another upper bound for S, it holds that $a \leq b$. If S has an supremum it is unique. Therefore we denote this supremum as $\sup S$.

A poset W is said to have the *least upper bound (LUB) property* if whenever a nonempty subset S has an upper bound in W, it has a least upper bound in W. The natural analogues of these quantities are *lower bounds, greatest lower bounds or infimums,* and the *greatest lower bound (GLB) property*. The notation for the infimum of the set S is $\inf S$.

The set $\mathbb{Q}$ of rationals does not have the LUB property, while $\mathbb{Z}$ and $\mathbb{N}$ do. In fact, $\mathbb{Z}$ and $\mathbb{N}$ have something stronger. Any non-empty subset S of $\mathbb{Z}$ or $\mathbb{N}$, that is bounded above, has its supremum in S itself. In order to show this we assume the following characterization of the integers.

Proposition 1.1. *$\mathbb{Z}$ is the smallest subring of $\mathbb{R}$ whose positive elements, say $\mathbb{Z}_+$, are well-ordered, that is, if $S \subset \mathbb{Z}_+$ and S is non-empty, then S has a minimum.*

The above also gives a characterization of the natural numbers as being the positive elements of $\mathbb{Z}$.

Corollary 1.1. *Suppose $s, t \in \mathbb{Z}$. Then, either $s \leq t$ or $s \geq t+1$.*

Proof. Suppose not, that is, suppose $s, t \in \mathbb{Z}$ such that $t < s < t+1$. Consider $r = s - t$. We have $r \in \mathbb{Z}$ and $0 < r < 1$. Let

$$S = \{r, r^2, \dots\}.$$

As $r > 0$, we have $r^n > 0$ for all $n \in \mathbb{N}$. Therefore, $S \subset \mathbb{Z}_+$. By the well-ordering principle, S has a minimum, say r^k. Further, as $r < 1$, we get $r^k r < r^k$ or $r^{k+1} < r^k$. This is a contradiction to the minimality of r^k. □

Proposition 1.2. *Let $S \subset \mathbb{Z}$ be non-empty such that S has an upper bound. Then $\sup S \in S$.*

Proof. We know that $\sup S \in \mathbb{R}$. Call this number α. Note that $\alpha - 1$ is not an upper bound for S, therefore there exists $s \in S$ such that

$$s \leq \alpha \quad \text{and} \quad \alpha - 1 < s.$$

Let $t \in S$. Suppose $t \geq s+1$. Then $\alpha < s+1 \leq t$, which is a contradiction, as α is an upper bound for S. Therefore, we have $t \leq s$. This shows that s is an upper bound for S, which in turn implies that $\alpha \leq s$. Since $\alpha \geq s$ also, we conclude that $\alpha = s$. □

Definition 1.1. The set of real numbers, denoted by $\mathbb{R}$, is the smallest ordered field containing $\mathbb{Q}$ (as an ordered field) that has the LUB property.

Using just this definition alone we can prove many familiar properties of $\mathbb{R}$. For $x \in \mathbb{R}$ let

$$U(x) = \{z \in \mathbb{Z} \mid z \geq x\} \text{ and } L(x) = \{z \in \mathbb{Z} \mid z \leq x\}.$$

Since $\mathbb{Z}$ has the least upper bound property, we can define the following $\mathbb{Z}$-valued functions on $\mathbb{R}$,

$$\lceil x \rceil = \inf U(x) \text{ and } \lfloor x \rfloor = \sup L(x).$$

These are the familiar *ceiling* and *floor* functions, respectively. Below we list some properties of these functions.

$\lfloor x \rfloor \leq x \leq \lceil x \rceil$	To see this, note that x is a lower bound for $U(x)$. So $$\lceil x \rceil = \inf U(x) \geq x.$$ The other side of the inequality can be similarly argued.
$0 \leq \lceil x \rceil - \lfloor x \rfloor \leq 1$	The left inequality follows from point 1. To see the right one, note that the integer $\lfloor x \rfloor + 1$ is not in $L(x)$, as $\lfloor x \rfloor$ is an upper bound for $L(x)$. Since $\mathbb{Z}$ has a total order, it must be that $\lfloor x \rfloor + 1 \geq x$. Therefore, $\lfloor x \rfloor + 1 \in U(x)$. This shows that $\lceil x \rceil \leq \lfloor x \rfloor + 1$, and $\lceil x \rceil$ is a lower bound for $U(x)$.

Proposition 1.3. *Between any two real numbers there is an irrational number.*

Proof. Suppose $a, b \in \mathbb{R}$ such that $a < b$. Let γ be any irrational number. We have $a - \gamma \leq b - \gamma$. We know that there exists $q \in \mathbb{Q}$ such that

$$a - \gamma \leq q \leq b - \gamma.$$

This implies

$$a \leq q + \gamma \leq b.$$

Now, $q + \gamma$ is not a rational number, because if it were, then γ would be a rational number. □

1.3 Exercises

Problem 1.1. Suppose U is an Abelian group. Let P be a subset of U with the following properties:

i. $0 \in P$,

ii. if $p \in P$ and $-p \in P$ then $p = 0$,

iii. if $p_1, p_2 \in P$ then $p_1 + p_2 \in P$.

Such a subset is called a *positive cone*. Use P to define a partial order on U.

Conclude (informally) that for an Abelian group, having a partial order is equivalent to having a positive cone.

Problem 1.2. [optional] Given $a, b \in \mathbb{R}$ such that $a < b$, give an algorithm to find a $q \in \mathbb{Q}$ such that $a < q < b$. Argue why your algorithm is correct.

Problem 1.3. Let F be an ordered field. Define the absolute value function $|\cdot|$ on F as follows:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}.$$

Show the following

1. $|x| \geq 0$, and $|x| = 0$ if and only if $x = 0$
2. $|x| = |-x|$
3. $x \leq |x|$
4. $|x + y| \leq |x| + |y|$
5. $|x| \leq a$ if and only if $-a \leq x \leq a$
6. $|x| \geq a$ if and only if $x \geq a$ or $x \leq -a$.

Give a one line reason why the function $d(x, y) = |x - y|$ is not a valid metric on F.

Problem 1.4. Suppose a non-zero polynomial p is expressed as $p(x) = p_m + p_{m-1}x + \cdots + p_0 x^m$ for some nonnegative integer m, where $p_0 \neq 0$. Let

$$F = \left\{ \frac{p}{q} : p, q \text{ polynomials and } q \neq 0 \right\}.$$

1. Show that F is a field.
2. Let $P = \{{}^{p}/_{q} \in F : p_0 q_0 \geq 0\}$. Show that P is a positive cone (refer to Problem 1.1). Conclude that F is a partially ordered field.
3. Show that F does not have the Archimedean property.

Problem 1.5. A theorem due to Hölder states that *every Archimedean ordered field is isomorphic*[1.1] *to a subfield of* $\mathbb{R}$. Use this to conclude that every Archimedean ordered field can be made into a metric space.

2 Sequences

2.1 Convergent and Cauchy sequences

Assuming the definition of a convergent sequence, let us look at some problems.

1.1. An isomorphism between two ordered fields is a bijection f that preserves the operations and the order, that is, $f(a \cdot b + c) = f(a) \cdot f(b) + f(c)$ and $f(a) \leq f(b)$ whenever $a \leq b$.

Problem 2.1. Let x_n be a sequence and x be a number. Suppose for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \frac{1}{n}$$

for all $n \geq N$. Is x_n convergent? Does this hold for every convergent sequence?

Answer 2.1. Yes, a sequence satisfying the conditions of the problem will be convergent, but not every convergent sequence will satisfy this condition. That is, the condition is sufficient for convergence, but not necessary.

However, one observes that showing that $|x_n - x|$ is eventually dominated by another nonnegative sequence that is going to 0, is sufficient for the convergence of x_n. This technique can often save the hassle of going back to the ε-N definition of convergence.

Proposition 2.1. *Show that a sequence of rational numbers can converge to an irrational number.*

Proof. We have shown earlier that between any two real numbers there is a rational number. Let α be any irrational number. Define x_n as any rational number picked between $\alpha - 1/n$ and α. Then

$$|x_n - \alpha| < \frac{1}{n} \quad \text{for all } n.$$

We have seen that this is a sufficient condition for convergence. □

Proposition 2.2. *Show that a sequence of irrational numbers can converge to a rational number.*

Proof. Consider the sequence $x_n = \sqrt{2}/n$. Clearly, x_n is an irrational number, because if it were not, then $\sqrt{2}$ would be rational. This sequence converges to the limit 0, as the sequence $1/n$ converges to the limit 0. □

Definition 2.1. A sequence $(x_n) \subset F$ is called a *Cauchy sequence*, if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $|x_m - x_n| < \varepsilon$ whenever $m \geq n \geq N$.

Textbooks generally write the condition $m, n \geq N$. But, nothing is lost by assuming $m \geq n$, because $|x_m - x_n| = |x_n - x_m|$. Let us illustrate the definition with a standard example.

Problem 2.2. Show that the sequence $x_n = \frac{1}{n}$ is Cauchy.

Proof. Note that for any $m \geq n$, we have

$$|x_m - x_n| \;=\; \frac{m-n}{mn} \;=\; \frac{1}{n}\frac{m-n}{m} \;\leq\; \frac{1}{n}\,.$$

Pick any $\varepsilon > 0$. Let $N = \lceil 1/\varepsilon \rceil + 1$. We get

$$\frac{1}{\varepsilon} < N \;\text{ or }\; \frac{1}{N} < \varepsilon.$$

So for any $m \geq n \geq N$, we get

$$|x_m - x_n| \leq \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Hence, the sequence is Cauchy. □

Problem 2.3. Let x_n be a sequence such that $|x_{n+1} - x_n| < 2^{-n}$ for all $n \in \mathbb{N}$. Show that x_n is Cauchy.

Proof. For any $m \geq n$, note that

$$\begin{aligned} |x_m - x_n| &= |x_m - x_{m-1} + x_{m-1} - x_{m-2} + \cdots + x_{n+1} - x_n| \\ &\leq |x_m - x_{m-1}| + \cdots + |x_{n+1} - x_n| \\ &\leq 2^{-n} + \cdots + 2^{-(m-1)} \\ &= 2^{-(n-1)}[2^{-1} + \cdots + 2^{-(m-n)}] \\ &\leq 2^{-(n-1)}. \end{aligned}$$

Pick any $\varepsilon > 0$. As the sequence $2^{-(n-1)}$ is convergent with the limit 0, there exists $N \in \mathbb{N}$ such that

$$2^{-(n-1)} < \varepsilon \quad \text{for all } n \geq N.$$

Therefore, for any $m \geq n \geq N$, we get

$$|x_m - x_n| \leq 2^{-(n-1)} < \varepsilon.$$ □

From this problem we extract the following proposition.

Proposition 2.3. *Suppose s_n is a sequence such that the series $\sum s_n$ converges absolutely. Let x_n be another sequence such that $|x_n - x_{n-1}| \leq |s_n|$ for all $n > 1$. Show that x_n is a Cauchy sequence.*

Proof. Define $S_n = \sum_{k=1}^{n} |s_k|$. The series $\sum s_n$ converging absolutely is the same as the sequence S_n converging. Therefore, S_n is also a Cauchy sequence. Now, note that for any $m > n$, we have

$$\begin{aligned} |x_{m+1} - x_{n+1}| &= |x_{m+1} - x_m + x_m - x_{m-1} + \cdots + x_{n+2} - x_{n+1}| \\ &\leq |x_{m+1} - x_m| + \cdots + |x_{n+2} - x_{n+1}| \\ &\leq |s_m| + \cdots + |s_{n+1}| \\ &= S_m - S_n \\ &\leq |S_m - S_n|. \end{aligned}$$

Choose any $\varepsilon > 0$. Since S_n is Cauchy, there exists $N \in \mathbb{N}$ such that

$$|S_m - S_n| < \varepsilon \quad \text{for all } m, n \geq N.$$

Therefore, whenever $m, n \geq N$, we also have

$$|x_m - x_n| \leq |S_m - S_n| < \varepsilon. \qquad \square$$

Note that absolute convergence of $\sum s_n$ is necessary. The series $\sum s_n$ where $s_n = 1/n(-1)^{n+1}$ is convergent but not absolutely convergent. The sequence of partial sums $S_n = \sum |s_n|$ satisfies $|S_n - S_{n-1}| = 1/n = |s_n|$. However, the sequence is not convergent.

Now, to take our understanding further, let us study a special sequence. Let

$$e_n = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}.$$

Note that $j! \geqslant 2^j$ for all $j \geq 4$. To see this we note that $4! \geq 2^4$, and from here we see inductively that

$$(j+1)! = (j+1)\, j! \geq (j+1)\, 2^j \geq 2 \cdot 2^j = 2^{j+1}.$$

Clearly, e_n is a monotonically increasing sequence. And the above relation gives

$$e_n \leq 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{2^4} + \cdots + \frac{1}{2^n} \leq 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{8} \leq 3.$$

We conclude that e_n is convergent. The famous number e is (often) *defined* as the limit of this sequence, that is,

$$e = \lim_{n \to \infty} e_n = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \cdots.$$

Problem 2.4. Show that $\lim \left(1 + \frac{1}{n}\right)^n = e$.

Proof. Let $x_n = \left(1+\frac{1}{n}\right)^n$. Use the binomial theorem to get

$$x_n = \sum_{j=0}^{n} \frac{p_{n,j}}{j!} \quad \text{where} \quad p_{n,j} = \prod_{k=0}^{j-1} \left(1-\frac{k}{n}\right),$$

with the understanding that $p_{n,0} = 1$. As an aside, since $(1 - {}^{k}/_{n})$ is incerasing in n, so is $p_{n,j}$ and, therefore, x_n. Further, we have $0 \le p_{n,j} \le 1$. This gives

$$x_n \le e_n < e.$$

Being monotonically increasing and bounded above, x_n is also convergent. One may intuit that e is the limit of x_n, noting that $\lim_{n\to\infty} p_{n,j} = 1$. However, the problem is that the limit cannot be exchanged with the sum, as number of terms in the sum itself grows with n. To circumvent this, we define a new function

$$P_n(m) = \sum_{j=0}^{m} \frac{p_{n,j}}{j!}.$$

Note that $x_n = P_n(n)$. Further note that for all $n \ge m$, we have

$$|x_n - e| \le |x_n - P_n(m)| + |P_n(m) - e|.$$

Now, choose any $\varepsilon > 0$. Since e_n converges to the limit e, there exists an m_0 such that

$$e_{m_0} > e - \frac{\varepsilon}{2}.$$

Indeed the inequality will hold for all $n \ge m_0$, but we focus only the term e_{m_0}. Further, since $p_{n,j}$ goes to 1 as n grows to infinity, we get that $P_n(m_0)$ converges to the limit e_{m_0}. Since m_0 is independent of n, this time the limit can easily be taken inside the sum. Due to this convergence, we know that there exists an $N \ge m_0$ such that

$$P_N(m_0) \ge e_{m_0} - \frac{\varepsilon}{2}.$$

Now we note that for all $n \ge N \ge m_0$, since x_n is an increasing function, we get

$$e > x_n \ge x_N = P_N(N) \ge P_N(m_0) \ge e_{m_0} - \frac{\varepsilon}{2} \ge e - \varepsilon.$$

The inequality $P_N(N) \ge P_N(m_0)$ is simply the consequence of truncating the sum $P_N(N)$ of N nonnegative numbers to just m_0 of them. The extreme sides of the obtained inequality gives

$$e - \varepsilon \le x_n < e \quad \text{for all } n \ge N.$$

This shows that the sequence x_n is convergent with the limit e. □

2.2 Limit superior and limit inferior

Let x_n be a sequence. The number L called the *limit superior* of x_n, denoted by $\limsup x_n$ or $\overline{\lim}\, x_n$, are defined in either of the following ways.

(I) For every $\varepsilon > 0$

(i) only a finitely many terms of x_n are larger than $L+\varepsilon$ and

(ii) infinitely many terms of x_n are strictly larger than $L-\varepsilon$.

Similarly, the *limit inferior* of x_n is a number L, denoted by $\liminf x_n$ or $\underline{\lim}\, x_n$, such that for every $\varepsilon > 0$

(i) only a finitely man terms of x_n are smaller than $L-\varepsilon$ and

(ii) infinitely many terms of x_n are strictly smaller than $L+\varepsilon$.

(II) Let $X_k = \{x_k, x_{k+1}, \dots\}$. Then, alternatively, we have

$$\limsup x_n = \lim_{k\to\infty} \sup X_k \text{ and } \liminf x_n = \lim_{k\to\infty} \inf X_k.$$

It is assumed here that tht underlying space has the LUB property, otherwise the infimum and supremum may not make sense. Unless mentioned otherwise, for this discussion, all sequences will be in $\mathbb{R}$. Now, note that $\sup X_k$ and $\inf X_k$ are decreasing and incrasing sequences respectively. Therefore, if we include $\pm\infty$ in our system, then every sequence x_n has a limit superior and a limit inferior.

(III) Let $\Lambda(x_n)$ be the set of all subsequential limits of x_n. That is, for every $l \in \Lambda(x_n)$, there exists a subsequence x_{n_j} of x_n such that x_{n_j} converges to l. Then, another definition of limit superior and limit inferior is as follows:

$$\limsup x_n = \sup \Lambda(x_n) \text{ and } \liminf x_n = \inf \Lambda(x_n).$$

Problem 2.5. Let x_n and y_n be two sequences. Is it true that

$$\limsup x_n y_n \leq_? (\limsup x_n)(\limsup y_n)?$$

Proof. Not necessarily. Consider the sequence

$$0, -1, 0, -1, \dots.$$

Let x_n and y_n be both this sequence. Then $\limsup x_n = 0$. So the right hand side of the inequality is 0. The sequence $x_n y_n$ is

$$0, 1, 0, 1, \dots.$$

Then the left hand side $\limsup x_n y_n = 1$. This is larger than the right hand side. □

Question 2.1. Let q_n be any enumeration of the rational numbers. What is $\Lambda(q_n)$? Conclude that a sequence may not only have a countably infinite set of subsequential limits, but also an uncountable one.

The following properties of limit superior and limit inferior have been seen.

1. $\liminf x_n \leq \limsup x_n$ for all x_n.
2. If x_n is convergent if andn only if $\limsup x_n = \liminf x_n = \lim x_n$.
3. A corollary of the point above is that every Cauchy sequence in $\mathbb{R}$ is convergent (earlier we had seen that every convergent sequence is Cauchy).
4. There is a subsequence of x_n that converges to $\limsup x_n$ and $\liminf x_n$. As a corollary we get Bolzano-Weierstrass theorem.
5. If $x_n \neq 0$, then
$$\limsup |x_n|^{\frac{1}{n}} \leq \limsup \left| \frac{x_{n+1}}{x_n} \right|.$$
For $\liminf$, the inequality goes in the other direction. This also tells us that if $|{}^{x_{n+1}}/_{x_n}|$ is convergent, then so is $|x_n|^{1/n}$, and to the same limit.

2.3 Exercises

Problem 2.6. Let x_n be a sequence. Consider the subsequences x_{2n} and x_{2n-1} of the terms of x_n at the even and odd indices, respectively. Suppose both these subsequences converge to the limit L. Using the (ε, N) definition, show that the sequence x_n also converges, and has the same limit L.

Problem 2.7. Let x_n be a Cauchy sequence. Let y_n be a subsequence of x_n such that y_n is convergent with the limit L. Show that x_n also converges to L.

Problem 2.8. Show that the definitions (I), (II) and (III) in section 2.2 are indeed equivalent.

Problem 2.9. Show that the supremum and the infimum in definition (III) in section 2.2 can be replaced by maximum and minimum, respectively.

Problem 2.10. Let $x_n = \frac{(-1)^n n}{(-1)^{n+1} + n}$. Calculate $\limsup x_n$ and $\liminf x_n$.

Problem 2.11. Let x_n be any convergent sequence. Define a new sequence σ_n where
$$\sigma_n = \frac{(x_1 + \cdots + x_n)}{n}.$$

Show that σ_n is also convergent and $\lim \sigma_n = \lim x_n$.

For the statements in coming problems **either give a proof or disprov**e them with a clear counterexample.

Problem 2.12. Let x_n be a bounded sequence. Then $\sup\{x_n : n \in \mathbb{N}\} = \limsup x_n$.

Problem 2.13. For every bounded sequence x_n there exists $\varepsilon > 0$ such that no term of x_n is smaller than $(\liminf x_n) - \varepsilon$.

Problem 2.14. Every term of a sequence x_n may be smaller than $\liminf x_n$.

Problem 2.15. If $\sup x_n = \infty$ then $\limsup x_n = \infty$.

Problem 2.16. For every sequence x_n we have $\limsup x_n = -\limsup(-x_n)$.

Problem 2.17. For every sequence x_n we have $\liminf x_n = -\limsup(-x_n)$.

Problem 2.18. Let x_n be a sequence and L be a number. Suppose for every $\varepsilon > 0$

i. there exists an $N \in \mathbb{N}$ such that $x_n < L + \varepsilon$ for all $n \geq N$,

ii. there exists an $m \in \mathbb{N}$ such that $x_m > L - \varepsilon$.

Then L must be $\limsup x_n$.

Problem 2.19. Let x_n be a sequence of **distinct** rational numbers. Then

$$\limsup \sin x_n = 1.$$

Problem 2.20. Suppose x_n is a bounded sequence such that it is **not** convergent. Then x_n has at least 2 subsequential limits.

3 Series

3.1 Convergence, absolute and conditional convergence

A series is an infinite sum of the terms of a sequence, usually denoted by $\sum x_n$. In the notation, the sum is just a formal symbol, and does not necessarily represent the sum of the series. When we say two series $\sum x_n$ and $\sum y_n$ are equal, we do not mean that their sum is equal (for the sum may not exist), but just that the sequence x_n and y_n are identical.

For any series $\sum x_n$ one can construct the seqeunce of its partial sums

$$s_m = \sum_{n=1}^{m} x_n.$$

Thus, one can talk about the convergence of a series by studying the convergence of the sequence of its partial sums. A series is called *convergent* if the sequence s_m of its partial sums is convergent, *absolutely convergent* if the positive-termed series $\sum |x_n|$ is convergent, and *conditionally convergent* if the series is convergent but not absolutely convergent.

One may be able to decide whether a series converges using tests like the comparison test, root test, ratio test and integral test.

Example 3.1. Show that if $a_n > 0$ and $\sum a_n$ is divergent, then so is $\sum \frac{a_n}{a_n+1}$.

Proof. Note that

$$\frac{a_n}{a_n+1} = \frac{1}{1+\frac{1}{a_n}} \leq \frac{\sqrt{a_n}}{2}.$$

If $\frac{a_n}{a_n+1} \not\to 0$, then there is nothing to prove. But if it is, then for n large enough and any $\varepsilon < \frac{1}{2}$,

$$\frac{a_n}{a_n+1} < \varepsilon \text{ or } a_n < \frac{\varepsilon}{1-\varepsilon} < 2\varepsilon$$

That is a_n goes to 0. Then eventually, $a_n < 1$ and $\frac{a_n}{a_n+1} \geq \frac{a_n}{2}$. The series diverges by comparison test. □

The following proposition gives a loose converse of the p-test. The p-test says that the series $\sum \frac{1}{n^p}$ converges if and only if $p > 1$.

Proposition 3.1. *Let $x_n \geq 0$ and monotonically decreasing such that the series $\sum x_n$ is convergent. Then $x_n = o(1/n)$.*

Proof. For any n consider the sum

$$n x_n \leq \sum_{k=n+1}^{2n} x_k.$$

By the Cauchy criterion of series convergence, the sum on the right hand side can be made arbitrarily small for large enough n. This shows that the sequence $n x_n$ is convergent to 0. Therefore, there exists M, such that

$$n x_n < M \text{ or } x_n < \frac{M}{n} \quad \text{for all } n.$$

Thus, $x_n = o(1/n)$. □

Proposition 3.2. *If a series $\sum x_n$ is absolutely convergent then it is convergent.*

Proof. Trivially we have $x_n \leq |x_n|$. For any $m \geq n$ we have

$$\left| \sum_{k=n}^{m} x_k \right| \leq \sum_{k=n}^{m} |x_k|.$$

Sine $\sum |x_n|$ is convergent, the right hand side can eventually be made arbitrarily small. So the series $\sum x_n$ converges by Cauchy criterion. □

Proposition 3.3. *Every rearrangement of an absolutely convergent series has the same sum.*

Proof. Let L be the sum of the series $\sum a_n$ and let $\sum b_n$ be any rearrangement of it. Then there exists a bijection f of $\mathbb{N}$ such that $a_n = b_{f(n)}$ for all n.

Choose any $\varepsilon > 0$. Since $\sum a_n$ is absolutely convergent, there exists a natural number N such that

$$\sum_{n=N+1}^{\infty} |a_n| < \frac{\varepsilon}{2}.$$

Let s_m and t_m be the sequence of partial sums of the series $\sum a_n$ and $\sum b_n$ respectively. Let $K = \max\{f(1), \ldots, f(N)\}$. For any $m \geq K$, consider the difference

$$|t_m - s_m| = \left| \sum_{n=1}^{M} b_n - \sum_{n=1}^{M} a_n \right|.$$

The terms in both the partial sums come from the set $\{a_k\}$. Any term a_k either appears in both the partial sums, in which case it gets cancelled, or it appears in at most one of the two partial sums. By our choice of m, the terms $a_1, \ldots, a_N$ are among those that appear in both and get cancelled. Therefore we conclude that

$$|t_m - s_m| < \sum_{n=N+1} |a_n| < \frac{\varepsilon}{2}.$$

Finally, since s_m is convergent to the limit L, we can pick $K_0 \geq K$ such that

$$|s_m - L| < \frac{\varepsilon}{2} \quad \text{for all } m \geq K_0.$$

Finally, for all $m \geq K_0$ we get

$$|t_m - L| \leq |t_m - s_m| + |s_m - L| < \varepsilon. \qquad \square$$

Proposition 3.4. [Riemann rearrangement theorem] *If the series $\sum x_n$ is conditionally convergent, then given any $\alpha \in \mathbb{R}$, there exists a rearrangement $\sum y_n$ of the series $\sum x_n$, such that $\sum y_n$ is convergent with the sum α.*

Proof. *A translation of Riemann's original proof*[3.1] *of the series rearrangement theorem: a conditionally convergent series can be rearranged to have any arbitrary sum.*

... infinite series fall into two distinct classes, depending on whether or not they remain convergent when all the terms are made positive. In the first class the terms can be arbitrarily rearranged; in the second, on the other hand, the value is dependent on the ordering of the terms.

Indeed, if we denote the positive terms of a series in the second class by $a_1, a_2, a_3, \ldots$ and the negative terms by $-b_1, -b_2, -b_3, \ldots$ then it is clear that $\sum a_n$ an as well as $\sum b_n$ must be infinite. For if they were both finite, the series would still be convergent after making all the signs the same. If only one were infinite, then the series would diverge.

Clearly now an arbitrarily given value α can be obtained by a suitable reordering of the terms. We take alternately the positive terms of the series until the sum is greater than α, and then the negative terms until the sum is less than α. The deviation from α never amounts to more than the size of the term at the last place the signs were switched. Now, since the numbers a_n as well as the numbers b_n become infinitely small with increasing index, so also are the deviations from α. If we proceed sufficiently far in the series, the deviation becomes arbitrarily small, that is, the series converges to α. $\square$

The reader is encouraged to absorb the quality of Riemann's proof, the economy of symbols, the use of language, and the clarity of the arguments.

Proposition 3.5. *If $\sum a_n$ and $\sum b_n$ are absolutely convergent, then so is $\sum a_n b_n$.*

Proof. Since $\sum b_n$ is absolutely convergent, $|b_n|$ goes to 0. In particular $|b_n|$ is bounded, that is, there exists M such that $|b_n| \leq M$ for all n. So,

$$\sum_{n=1}^{\infty} |a_n b_n| \leq M \sum_{n=1}^{\infty} |a_n|.$$

3.1. Riemann, Bernhard (1868). "Über die Darstellbarkeit einer Function durch eine trigonometrische Reihe". *Abhandlungen der Königlichen Gesellschaft der Wissenschaften zu Göttingen.* **13**: 87–132.

The partial sums of $\sum |a_n\, b_n|$ are increasing and bounded, and hence the sum converges. Hence $\sum a_n\, b_n$ converges absolutely. □

3.1.1 Exercises

Problem 3.1. Consider the series $\sum x_n$ where $x_n = \frac{2^n}{n!}$ for $n \geq 0$.

a. Prove that the series $\sum x_n$ is convergent.

b. Show that $2 \cdot 3^{n-2} \leq n!$ for all $n \geq 2$.

c. Use this to show that $\sum_{n=0}^{\infty} x_n \leq 9$.

Problem 3.2. Consider the series $\sum x_n$ where $x_n = \log\left(1 - \frac{(-1)^n}{n}\right)$ for $n \geq 1$.

a. Show that $\sum x_n$ is an alternating series, that is, the terms of the sequence x_n are alternatingly positive and negative.

b. Explain why the series $\sum x_n$ is convergent.

c. Find the sum $\sum_{n=1}^{\infty} x_n$.

Problem 3.3. Suppose $a_n > 0$ and the series $\sum a_n$ converges. Show that the series $\sum \frac{a_n}{1+a_n}$ is also convergent.

Problem 3.4. Suppose the seqeunce $a_n \geq 0$ for all $n \in \mathbb{N}$, and the series $\sum a_n$ is convergent. Then show that the series $\sum \frac{\sqrt{a_n}}{n}$ is convergent.

Problem 3.5. Determine whether the following statements are true or false. Give either a proof or a counterexample for each.

1. If the series $\sum a_n$ converges according to the ratio test, then it converges according to the root test.
2. If the series $\sum a_n$ diverges according to ratio test, then it diverges according to the root test.
3. The series $\sum a_n$ may converge according to the root test but diverge according to the ratio test.
4. If the series $\sum a_n$ and $\sum b_n$ are convergent, then so is the series $\sum a_n\, b_n$.
5. If $\sum a_n\, b_n$ is convergent then both the series $\sum a_n$ and $\sum b_n$ must be convergent.
6. If $\sum a_n\, b_n$ and $\sum a_n$ is convergent, then so is $\sum b_n$.
7. If $\sum a_n\, b_n$ is convergent then at least one of the two series $\sum a_n$ and $\sum b_n$ must be convergent.

8. If $\sum a_n b_n$ is convergent and $a_n, b_n \geq 0$, then both the series $\sum a_n$ and $\sum b_n$ must be convergent.

3.2 Convergence of the series $\sum a_n b_n$

Let a_n and b_n be two sequences. We have seen some results on this theme: conditions on the series $\sum a_n$ or $\sum b_n$, or the sequences a_n and b_n might enforce the convergence of the series $\sum a_n b_n$. We will see some more results in this vein.

The first reasonable thing to try is whether the convergence of the series $\sum a_n$ and $\sum b_n$ enforces that of $\sum a_n b_n$. We have seen that this is not true. However, if $\sum a_n$ and $\sum b_n$ are absolutely convergent, then so is the series $\sum a_n b_n$. One may have noticed in the proof that the requirement of absolute convergence on both the series was an overkill. If $\sum a_n$ is absolutely convergent and the sequence b_n is bounded, then the series $\sum a_n b_n$ is absolutely convergent. This is a big relaxation on b_n, while the condition on $\sum a_n$ remains strict. On the other hand, the conclusion is also more than what we need.

If $\sum a_n$ is convergent, and the sequence b_n is convergent, then does the series $\sum a_n b_n$ converge? The answer is no again, as the stronger condition of $\sum b_n$ converging does not suffice. If we enforce $a_n \geq 0$, and $\sum a_n$ to be convergent, then we do get $\sum a_n b_n$ converging absolutely. But we have said nothing new, for the condition on a_n is like the seen case of $\sum a_n$ being absolutely convergent. If we demand $\sum a_n$ to just be convergent, what additional condition on the sequence b_n might enforce the convergence of the series $\sum a_n b_n$? The natural first guess is $b_n \geq 0$. This does not suffice, and the example is clever. We present it in a coming proposition. However, it turns out that imposing monotonicity on b_n (along with convergence) sufficies for $\sum a_n b_n$ to converge. This result is due to Abel and is presented in a coming theorem.

Sr.	Conditions on a_n and b_n	Conseq.
1.	$\sum a_n, \sum b_n$ both convergent	$\times$
2.	$\sum a_n, \sum b_n$ both absolutely convergent	$\checkmark +$ a.c.
3.	$\sum a_n$ absolutely convergent, b_n bounded	$\checkmark +$ a.c.
4.	$\sum a_n$ convergent, b_n convergent	$\times$
5.	$\sum a_n$ convergent, $a_n \geq 0$, b_n convergent	$\checkmark +$ a.c.
6.	$\sum a_n$ convergent, b_n convergent and $b_n \geq 0$	$\times$
7.	$\sum a_n$ convergent, b_n convergent and monotone	$\checkmark$

Table 3.1. Summary of conditions on the sequences a_n and b_n that are sufficient (or insufficient) to ensure convergence of $\sum a_n b_n$ (indicted in the rightmost column).

[a.c. = absolutely convergent]

Proposition 3.6. *There are sequences a_n, b_n such that $b_n \geq 0$ and $\sum a_n$ converges, but the series $\sum a_n b_n$ diverges.*

Proof. Let $a_n = \frac{(-1)^{n+1}}{\sqrt{n}}$. The series $\sum a_n$ converges by the alternating series theorem. Let

$$b_n = \begin{cases} \frac{1}{\sqrt{n}}, & n \text{ odd} \\ \frac{1}{n\sqrt{n}}, & n \text{ even.} \end{cases}$$

The sequence b_n has the required properties. The sequence $a_n b_n$ then is as follows.

$$a_n b_n = \begin{cases} \frac{1}{n}, & n \text{ odd} \\ -\frac{1}{n^2}, & n \text{ even.} \end{cases}$$

Consider the partial sums s_N of the series $\sum a_n b_n$. We have

$$s_{2N} = \sum_{n=1}^{N} \frac{1}{2n-1} - \sum_{n=1}^{N} \frac{1}{4n^2} > \sum_{n=1}^{N} \frac{1}{2n-1} - \frac{1}{4}\sum_{n=1}^{\infty} \frac{1}{n^2}.$$

The odd harmonic series diverges, so s_{2N} grows to infinity. □

Theorem 3.1. *Let a_n and b_n be sequences such that $\sum a_n$ is convergent, and the sequence b_n be convergent and monotone. Then $\sum a_n b_n$ is convergent.*

Proof. Let s_n be the sequence of partial sums of $\sum a_n$ with the convention that $s_0 = 0$. Then $a_n = s_n - s_{n-1}$ for all $n \geq 1$. Let t_n be the sequence of partial sums of $\sum a_n b_n$. Then

$$t_N = \sum_{n=1}^{N} a_n b_n = \sum_{n=1}^{N} (s_n - s_{n-1}) b_n = \sum_{n=1}^{N} s_n b_n - \sum_{n=1}^{N} s_{n-1} b_n.$$

Reindex the second sum by replacing $n-1$ by n (with $s_0 = 0$). We get

$$t_N = s_N b_N + \sum_{n=0}^{N-1} s_n(b_n - b_{n+1}).$$

Let r_N denote the partial sum in the above expression. Let us consider the series $\sum (b_n - b_{n+1})$. In particular

$$\sum_{n=0}^{m-1} |b_n - b_{n+1}| = |b_0 - b_m| < 2 \sup_n |b_n| < \infty.$$

This series in question is thus absolutely convergent. Further, the seqeunce s_n is convergent (and hence bounded), So the sequence of partial sums r_N is also convergent. Thus we have expressed t_N as a sum of two convergent sequences $s_N b_N$ and r_N. Therefore, t_N is convergent. □

3.3 Cauchy products of sequences and series

Definition 3.1. Let a_n and b_n be two sequences indexed from $n=0$. We define a new sequence $a_n * b_n$ as follows. Let $c_n = a_n * b_n$. Then

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}.$$

The sequence c_n is called the *Cauchy product* of the sequences a_n and b_n. Similarly, for two series $\sum a_n$ and $\sum b_n$, their Cauchy product is the series $\sum (a_n * b_n)$.

Loosely speaking, on the space of sequences, the product $*$ has an identity, namely the sequence $1, 0, 0, \ldots$. Further, the product $*$ is commutative and it distributes over sum, that is,

$$(a_n + b_n) * p_n = (a_n * p_n) + (b_n * p_n).$$

Turns out that the Cauchy product of two sequences to converge, one must impose strict conditions on the sequences. Both the sequences being convergent, or even summable, does not ensure the convergence of their Cauchy product. However, the coming proposition gives a sufficient condition.

Proposition 3.7. *Let a_n and b_n be two sequences. Suppose the series $\sum a_n$ is absolutely convergent, and its suppose its sum is α. Further, suppose the sequence b_n converges b. Then the sequence $a_n * b_n$ is convergent, and its limit is αb.*

Proof. Let us first prove this in the special case when $b = 0$. Let $c_n = a_n * b_n$. Let γ be the sum of the series $\sum |a_n|$. Choose any $\varepsilon > 0$. Since b_n converges to 0, there is some N such that $|b_n| < \varepsilon$. for all $n \geq N$. Further, by the Cauchy criterion of series convergence, there exists $M \geq N$ such that for all $n \geq m \geq M$

$$\sum_{k=m}^{n} |a_k| < \varepsilon.$$

Then, for all $n \geq M + N$

$$\begin{aligned} |c_n| &\leq |b_0|\,|a_n| + \cdots + |b_N|\,|a_{n-N}| + |b_{N+1}|\,|a_{n-(N+1)}| + \cdots + |b_n|\,|a_0| \\ &\leq |b_0|\,|a_n| + \cdots + |b_N|\,|a_{n-N}| + \varepsilon\gamma \\ &\leq \mu\,(|a_{n-N}| + \cdots + |a_n|) + \varepsilon\gamma \\ &\leq (\mu + \gamma)\varepsilon, \end{aligned}$$

where $\mu = \sup |b_n|$. Since ε are arbitrary, this shows that c_n converges to 0. Now, if b was arbitrary, write

$$c_n = a_n * [(b_n - b) + b] = a_n * (b_n - b) + a_n * b.$$

The sequence $b_n - b$ goes to 0. And $a_n * b = A_n b$. Here we identify the scalar b with the seqeunce $b, b, b, \ldots$. Taking limits, we get c_n tending to αb. □

Theorem 3.2. *Let a_n be $\sum a_n$ is absolutely convergent and $\sum_{n=0}^{\infty} a_n = A$. Let b_n be such that $\sum b_n = B$. Then their Cauchy product $\sum c_n$ is convergent and its sum is AB.*

Proof. Let A_n, B_n and C_n be the partial sums of $\sum a_n$, $\sum b_n$ and $\sum c_n$ respectively. Then

$$\begin{aligned} C_n &= a_0 b_0 + (a_0 b_1 + b_0 a_1) + \cdots + (a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0) \\ &= a_0 B_n + a_1 B_{n-1} + \cdots + a_n B_0 \\ &= a_n * B_n. \end{aligned}$$

Clearly the series a_n and B_n satisfy the conditions of proposition 3.7. Hence, C_n goes to AB. □

The conditions in the above proposition cannot be weakened to convergence of both series. To see this consider the seqeunces

$$a_n = b_n = \frac{(-1)^n}{\sqrt{2(n+1)}} \quad \text{for } n \geq 0.$$

The series $\sum a_n$ is convergent and the sequence b_n goes to 0. But note that

$$|c_n| = |a_n * b_n| = \left| \sum_{k=0}^{n} a_k b_{n-k} \right| = \left| (-1)^n \sum_{k=0}^{n} \frac{1}{\sqrt{2(k+1)}\,\sqrt{2(n-k+1)}} \right| \geq \sum_{k=0}^{n} \frac{1}{n+2},$$

where the last inequality follows from the arithmetic-geometric mean inequality. Clearly, the sequence c_n diverges.

Proposition 3.8. *The Cauchy product of two absolutely convergent series is absolutely convergent.*

Proof. Let $\sum a_n$ and $\sum b_n$ be absolutely convergent. Let $\sum c_n$ be their Cauchy product. Then

$$\sum_{n=0}^{N} |c_n| \leq \sum_{n=0}^{N} \sum_{k=0}^{n} |a_k|\,|b_{n-k}| \leq \sum_{n=0}^{N} |a_n| \sum_{n=0}^{N} |b_n| \leq \sum_{n=0}^{\infty} |a_n| \sum_{n=0}^{\infty} |b_n|,$$

where the second inequality relies on the fact that the terms of the sum on the left hand of it are contained in the terms of the sum on the right, and no term is negative. Again, the partial sums are increasing and bounded, and hence the series is convergent. □

Theorem 3.3. *Suppose a_n and b_n be such that $\sum a_n, \sum b_n$ and $\sum (a_n * b_n)$ are all summable. Then*

$$\sum_{n=0}^{\infty} (a_n * b_n) = \left(\sum_{n=0}^{\infty} a_n\right)\left(\sum_{n=0}^{\infty} b_n\right).$$

3.3.1 Exercises

Problem 3.6. Give examples of the following.

1. Convergent seqeunces a_n and b_n such that $a_n * b_n$ is not convergent.
2. Summable sequences a_n and b_n such that $a_n * b_n$ is not convergent.

4 Metric spaces and continuity

4.1 Metric Spaces

In the context of metric spaces we have looked at metrics, norms, limit points etc. We have looked at set operations like closure and interior. We have also looked at different set kinds like open, closed, connected, compact etc. Combining such sets using finite and infinite unions and intersections is a matter with interesting observations.

For a map $f: X \to Y$ where X and Y are metric spaces, we have looked at the image operator (also denoted by f) which carries a subset of X to a subset of Y; likewise, the pre-image operator f^{-1} (distinct from the inverse map f^{-1} when defined) taking a subset of Y to a subset of X. The pre-image operator is very "well-behaved".

4.1.1 Exercises

Problem 4.1. Determine whether the following statements are true or false. Give justification for each.

1. A set E is compact if every open cover of E that is made of infinitely many open sets has a finite subcover.
2. Let S be everywhere dense in $\mathbb{R}$. Suppose S is not open. Then $S' = \mathbb{R} - S$ is everywhere dense in $\mathbb{R}$.

Problem 4.2. In the space $\mathbb{R}^2$ determine whether the following sets are open/closed/neither open nor closed.

i. $\{(x, y) : xy \neq 0\}$

ii. $\{(x, y) : x^2 + y^2 = \pi\}$

iii. $\{(x, y) : y = x + 17\}$

iv. $\{(x, y) : y = x + 17e^x\}$

Problem 4.3. Let (Y, d_Y) be a metric space. Let X be any set such that there is an injection $f: X \to Y$. Define the function $d_X: X \times X \to \mathbb{R}$ by

$$d_X(x_1, x_2) = d_Y(f(x_1), f(x_2)).$$

Show that d_X is a metric on X.

Problem 4.4. Consider $f: X \to Y$ where X, Y are metric spaces.

1. Let $A \subseteq X$. Is one of $f(A')$ and $f(A)'$ always contained in the other? (Here S' is the complement of S in the corresponding metric space.)
2. Repeat part 1 when f is surjective.
3. Repeat part 1 when f is injective.
4. Let $A, B \subseteq X$. Show that $f(A \cap B) \subseteq f(A) \cap f(B)$.
5. Find an example where the containment in part 4 is strict.
6. Show that if f is injective then the containment in part 4 can be replaced by an equality.

Optional exercise.

Problem 4.5. Let x_n be any sequence such that $x_n \geq 0$. Let q_n be an enumeration of all the rational numbers. Define the family $\mathcal{Q}$ of open sets as follows

$$\mathcal{Q}(q_n, x_n) = \{(q_n - x_n, q_n + x_n)\}_{n=1}^{\infty}.$$

Questions:

1. Does there exist an enumeration q_n such that $\mathcal{Q}\left(q_n, \frac{1}{n}\right)$ is an open cover of $\mathbb{R}$?
2. Does there exist an enumeration q_n such that $\mathcal{Q}\left(q_n, \frac{1}{n}\right)$ is not an open cover of $\mathbb{R}$, that is, there are real numbers which are not in any open set in the family?
3. Repeat part 1. and part 2. with $x_n = \frac{1}{n^2}$.

What is your finding from this exercise? Can you generalize your findings into a conjecture? If so, try to prove it.

4.2 Continuity

Consider the map $f: X \to Y$ where X, Y are metric spaces equipped with metrics d_X and d_Y respectively. Then f is said to be *continuous at the point x_0 of X*, if either of the following holds:

I. (The ε-δ definition) For every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$f(S_\delta(x_0)) \subseteq S_\varepsilon(f(x_0)).$$

II. Whenever x_n is a convergent sequence in X with the limit x_0, the sequence $f(x_n)$ is convergent in Y with the limit $f(x_0)$.

The map f is said to be *continuous on X* if f is continuous at every point x_0 of X.

Theorem 4.1. *A map $f: X \to Y$ is continuous on X if and only if f pulls back open sets to open sets, that is, whenever E is an open set in Y, the set $f^{-1}(E)$ is open in X.*

The table below lists some properties which are (and are not) necessary and sufficient for a map to be continuous.

	A. Necessary property of continuous f		B. Sufficient for f to be continuous	
	i. Pulls back	ii. Preserves	i. Pulls back	ii. Preserves
(1) Open	✓	× const. on $\mathbb{R}$	✓	× $f = g^{-1}$ where $g: t \longmapsto e^{it}$ on $[0, 2\pi)$
(2) Closed	✓	× $\frac{1}{x}$ on $[1, \infty)$	✓	× sgn on $[-1, 1]$
(3) Bounded	× const. on $\mathbb{R}$	× $\frac{1}{x}$ on $(0, 1)$	× sgn on $[-1, 1]$	× sgn on $[-1, 1]$
(4) Compact	× const. on $\mathbb{R}$	✓	× $f(x) = \begin{cases} \frac{1}{x}, & x \in (0, 1] \\ 0, & x = 0 \end{cases}$	× sgn on $[-1, 1]$
(5) Connected	× x^2 on $\mathbb{R}$ $f^{-1}([1, 4])$ discon.	✓	× sgn on $[-1, 1]$	× $f(x) = \begin{cases} \sin\frac{1}{x}, & x > 0 \\ 0, & x = 0 \end{cases}$

Table 4.1. Continuity and the behavior of sets under images and preimages

Proposition 4.1. *Let $f: X \to Y$. Then f is continuous if and only if $f(\bar{A}) \subseteq \overline{f(A)}$ for all $A \subseteq X$.*

The proposition says that continuous functions are exactly those that preserve limit points of every set, that is, under and only under a continuous function the image of every limit point of any subset of X is a limit point of the image of the set (or in the image of the set).

Proof. Suppose f is continuous. Let $y \in f(\bar{A})$. Then, by definition, there exists a convergent sequence $x_n \in A$ and $x \in X$ such that $x = \lim x_n$ and $y = f(x)$. Since f is continuous, the sequence $f(x_n)$ whose terms are in $f(A)$ must also converge to y. Thus, $y \in \overline{f(A)}$.

Conversely, suppose $f(\bar{A}) \subseteq \overline{f(A)}$ for every $A \subseteq X$, but f is not continuous at some $x_0 \in X$. Thus, there exists a positive number ε such that for every δ there is a point within δ-distance of x_0 whose image under f is not within ε-distance of $f(x_0)$. Therefore, for each n, there is $x_n \in X$ such that

$$d_X(x_n, x_0) < \frac{1}{n} \text{ but } d_Y(f(x_n), f(x_0)) \geq \varepsilon.$$

Let $A = \{x_1, x_2, \dots\}$. Then $x_0 \in \bar{A}$, and by our hypothesis, $f(x_0) \in \overline{f(A)}$. But $f(x_0)$ is at least ε-distance away from every point of $f(A)$, and hence cannot be in $\overline{f(A)}$. This is a contradiction. □

Example 4.1. Consider $f: (0, \infty) \to \mathbb{R}$ where $f(x) = \frac{1}{x}$. Let $A = (0, \infty)$. Then $f(\bar{A}) = (0, \infty)$. But $\overline{f(A)} = [0, \infty)$. This shows that the containment condition in Proposition 4.1 may be strict.

Problem 4.6. Show that $f: X \to Y$ is continuous on X if and only if for all $A \subseteq Y$ we have

$$f^{-1}(\operatorname{Int} A) \subseteq \operatorname{Int} f^{-1}(A).$$

Example 4.2. The inverse of a continuous bijection may not be continuous. Let S be the unit circle in $\mathbb{C}$. Let $f: [0, 2\pi) \to S$ such that $f(x) = e^{ix}$. Then f is a continuous bijection. In fact, f is also uniformly continuous. To see this note that

$$|f(x) - f(y)| = |e^{ix} - e^{iy}| \leq |\cos x - \cos y| + |\sin x - \sin y| \leq 2\,|x - y|.$$

However, f^{-1} is discontinuous on the complex number 1.

Proposition 4.2. *Let $f: X \to Y$ be a continuous bijection where X is some compact subset of $\mathbb{R}^n$. Then f^{-1} is continuous.*

Proof. Suppose f^{-1} is not continuous at the point y. Then there exists a positive number ε and a convergent sequence y_n in Y converging to y, such that each $f^{-1}(y_n)$ is at least at a distance ε from $f^{-1}(y)$. Let $x_n = f^{-1}(y_n)$ and let $x = f^{-1}(y)$. Since X is compact, x_n has a convergent subsequence, say x_m, whose limit, say x_0, is in X. Since f is continuous, the sequence $f(x_m)$ is convergent with the limit $f(x_0)$. But $f(x_m)$ is a subsequence of the convergent sequence y_n and hence must share its limit, that is, $f(x) = f(x_0)$ or $x = x_0$. But this contradicts our hypothesis that each x_n is at least ε-distance away from x. This is a contradiction.

Alternate proof.

Let E be any closed subset of X. Since X is compact, so is E. Since f is continuous, $f(E)$ is compact, and hence closed. But $f(E) = (f^{-1})^{-1}(E)$. Thus we have shown that f^{-1} pulls back closed sets in X to closed sets in Y. Hence, f^{-1} is continuous. □

Proposition 4.3. *Let I be an interval and $f: I \to \mathbb{R}$ such that f preserves both compact sets and connected sets. Then f is continuous.*

Proof. Suppose f is not continuous at the point x. Let x_n be a monotone[4.1] sequence converging to x such that each $f(x_n)$ is at least ε-distance away from $f(x)$, for some $\varepsilon > 0$. Suppose $f(x_n)$ does not take any value infinitely often. Then $f(x_n)$ has a strictly monotone subsequence, say $f(x_m)$.

Note that the set $\{x_m\} \cup \{x\}$ is compact and so is its image $\{f(x_m)\} \cup \{f(x)\}$. Since $f(x_m)$ is monotone and bounded it must have a limit. This limit cannot be in $\{f(x_m)\}$, since the sequence is strictly monotone. Nor can $f(x_m)$ converge to $f(x)$, since each $f(x_m)$ is ε-distance away. This contradicts the compactness of $\{f(x_m)\} \cup \{f(x)\}$.

So, $f(x_n)$ must take a value c infinitely often. Without loss of generality assume $f(x_n) = c$ for all n (we can replace x_n by its subsequence that is constant) and $f(x) > c$ (if not, replace f by $-f$). Suppose x_n is monotonically increasing. Then, since f preserves connected sets, $f([x_n, x])$ contains the interval $[c, f(x)]$. So, for each n choose $y_n \in [x_n, x]$ such that $f(y_n) = c + \frac{f(x)-c}{n+1}$.

Now, note that $|y_n - x| < |x_n - x|$. So the sequence y_n converges to x. Therefore, the set $\{y_n\} \cup \{x\}$ is compact and so is its image $\{f(y_n)\} \cup \{f(x)\}$. By construction of y_n, the number c is a limit point of this set, but is not in the set as neither $f(y_n)$ nor $f(x)$ is equal to c. This is again a contradiction to the compactness of $\{f(y_n)\} \cup \{f(x)\}$.

Finally, if x_n is monotonically decreasing, then the same argument holds by replacing the interval (x_n, x) by (x, x_n). □

We collate our observations into the following theorem.

4.1. Such x_n can always be chosen since every sequence has a monotone subsequence.

Theorem 4.2. *Let $f: I \to \mathbb{R}$, where $I \subseteq \mathbb{R}$ is an interval (possibly infinite). Then the following are equivalent.*

I. The map f is continuous on I.

II. For every convergent sequence x_n in I with a limit x_0, the sequence $f(x_n)$ is convergent in $\mathbb{R}$ with the limit $f(x_0)$.

III. The map f pulls back open sets in $\mathbb{R}$ to open sets in I.

IV. The map f pulls back closed sets in $\mathbb{R}$ to closed sets in I.

V. For every $A \subseteq I$, we have $f(\bar{A}) \subseteq \overline{f(A)}$.

VI. For every $B \subseteq \mathbb{R}$, we have $f^{-1}(\operatorname{Int} B) \subseteq \operatorname{Int} f^{-1}(B)$.

VII. The image of every compact set under f is compact, and the image of every connected set under f is connected.

4.3 Uniform continuity

A function $f: X \to Y$ is called uniformly continuous if for every $\varepsilon > 0$, there exists $\delta > 0$ such that whenever $x, y \in X$ are such that $d_X(x, y) < \delta$, we have $d_Y(f(x), f(y)) < \varepsilon$.

For a continuous function f on X, given an ε, the number δ that witnesses the continuity of f at a point x_0 may depend on the point x_0 (as well as on ε). Uniform continuity demands the existence of a single δ that witnesses the continuity of f at every point of X.

Example 4.3. Continuous functions are exactly those that preserve convergent sequences, however, they need not preserve Cauchy sequences. Consider $f(x) = \frac{1}{x}$ on $(0, \infty)$. Let $x_n = \frac{1}{n}$ which is a Cauchy sequence in $(0, \infty)$. However, $f(x_n) = n$ is not Cauchy. A continuous function need not preserve Cauchy sequences.

Proposition 4.4. *A uniformly continuous function preserves Cauchy sequences.*

Proof. Let $f: X \to Y$ be uniformly continuous. Let x_n be any Cauchy sequence in X. Pick any $\varepsilon > 0$. Since f is uniformly continuous, there exists $\delta > 0$, such that $|f(x) - f(y)| < \varepsilon$ whenever $|x - y| < \delta$. Since x_n is Cauchy, there exists $N \in \mathbb{N}$ such that $|x_m - x_n| < \delta$ whenever $m, n \geq N$. Thus, whenever $m, n \geq N$, we have $|f(x_m) - f(x_n)| < \varepsilon$. Thus $f(x_n)$ is Cauchy in Y. □

Remark 4.1. Not every continuous map that preserves Cauchy sequences is uniformly continuous. Take any continuous function that is not uniformly continuous on a complete space, for example, $f(x) = x^2$ on $\mathbb{R}$. On $\mathbb{R}$, the Cauchy sequences are exactly the convergent sequences. Since f preserves convergent sequences, it also preserves Cauchy sequences.

Similarly, a map f that preserves Cauchy sequences must be continuous. To see this suppose x_n is any convergent sequence in X with the limit x. Then the sequence $x_1, x, x_2, x, \ldots$ is Cauchy and hence so is $f(x_1), f(x), f(x_2), f(x), \ldots$. This sequence has a limit in the completion of Y, but since the constant sequence $f(x)$ is a subsequence, this limit must be $f(x)$. So, we get

$$\begin{array}{ccccc} \text{Uniformly} & \Rightarrow & \text{Cauchy sequence} & \Rightarrow & \\ \text{continuous} & \not\Leftarrow & \text{preserving map} & \not\Leftarrow & \end{array} \quad \text{Continuous} \quad = \quad \begin{array}{c} \text{Convergent seq.} \\ \text{preserving map} \end{array} .$$

Corollary 4.1. *If $f\colon (a,b] \to \mathbb{R}$ is uniformly continuous, then $\lim_{x\to a^+} f(x)$ exists.*

Proof. Let x_n be any sequence in $(a,b]$ that is converging to a. Then x_n is Cauchy in $(a,b]$, and $f(x_n)$ Cauchy in $\mathbb{R}$. Therefore, $f(x_n)$ converges. Let y_n be another such sequence in $(a,b]$. Then the sequence $x_1, y_1, x_2, y_2, \ldots$ also converges to a, and hence the sequence $f(x_1), f(y_1), f(x_2), f(y_2), \ldots$ is convergent. And hence $f(y_n)$ has the same limit as $f(x_n)$. We conclude that $\lim_{x\to a^+} f(x)$ exists. □

Proposition 4.5. *Suppose $f\colon [a,\infty) \to \mathbb{R}$ is continuous and $\lim_{x\to\infty} f(x)$ exists. Then f is uniformly continuous on $[a,\infty)$.*

Proof. Choose any $\varepsilon > 0$. Suppose $L = \lim_{x\to\infty} f(x)$. Then there exists $b \geq a$ such that $|f(z) - L| < \frac{\varepsilon}{4}$ whenever $z \geq b$. Since f is continuous on $[a,b]$, it is also uniformly continuous. Hence, there exists $\delta > 0$ such that whenever $x, y \in [a,b]$ and $|x-y| < \delta$, we have $|f(x) - f(y)| < \frac{\varepsilon}{2}$.

Now, suppose $|x-y| < \delta$ and without loss of generality assume $x > y$. If $x \leq b$, we already have $|f(x) - f(y)| < \frac{\varepsilon}{2} < \varepsilon$. Let $x > b$. Then

$$|f(x) - f(y)| \leq |f(x) - f(b)| + |f(b) - f(y)|.$$

Observe that for any $z \geq b$ we have $|f(z) - f(b)| \leq |f(z) - L| + |f(b) - L| < \frac{\varepsilon}{2}$. So, $|f(x) - f(b)| < \frac{\varepsilon}{2}$. Now either $x > y \geq b$, in which case $|f(y) - f(b)| < \frac{\varepsilon}{2}$. Or $y < b < x$, in which case $|y - b| < \delta$ and again $|f(y) - f(b)| < \frac{\varepsilon}{2}$. Thus, in every case the two terms in the right hand side of the displayed inequality are smaller than $\frac{\varepsilon}{2}$. Hence $|f(x) - f(y)| < \varepsilon$. □

4.3.1 Exercises

Problem 4.7. Find examples of the following with necessary justification. You may use the internet.

1. A function on $\mathbb{R}$ that is discontinuous everywhere.
2. A function on $\mathbb{R}$ that is continuous at exactly one point.

3. A function on $\mathbb{R}$ that is continuous exactly at the integers.
4. A function on $\mathbb{R}$ that is continuous on the irrational numbers but discontinuous on the rationals.

Problem 4.8. Let $\mathbb{N}, \mathbb{Q}$ and $\mathbb{R}$ be equipped with the usual metric.

1. Show that every $f: \mathbb{N} \to \mathbb{R}$ is continuous.
2. Is such an f also uniformly continuous?
3. What about continuity of functions $f: \mathbb{Q} \to \mathbb{R}$?
4. On any subset X of $\mathbb{R}$, the function sgn is defined as $\operatorname{sgn}(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$. Is sgn continuous on $X = \left[-1, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, 1\right]$?

Problem 4.9. Let $f: X \to Y$ be a continuous bijection such that $X, Y \subseteq \mathbb{R}$.

1. Is f^{-1} necessarily continuous?
2. Is f^{-1} necessarily continuous if $X = Y = \mathbb{R}$?

Problem 4.10. Justify whether the following functions are uniformly continuous on $\mathbb{R}$

1. $f(x) = \sin x$,
2. $f(x) = x \sin x$,
3. $f(x) = \begin{cases} \sin 1/x, & x \neq 0 \\ 0, & x = 0 \end{cases}$,
4. $f(x) = \begin{cases} x \sin 1/x, & x \neq 0 \\ 0, & x = 0 \end{cases}$.